

The primary objective of this project is to develop a machine learning model capable of predicting "**Crowd Energy**" based on various features from the tour logs

1. Data Quality Assessment

Upon initial inspection, the dataset presented several data quality issues categorized as "Dirty" (quality) and "Messy" (structural) data.

1.1 Identified Issues

- **Inconsistent Date Formats (Show_DateTime):** The date and time information was recorded in multiple formats
- **Currency & Formatting Inconsistencies (Ticket_Price):** Prices were mixed with different currency symbols (£, €, \$, USD), contained text (e.g., "Free")
- **Missing Values:** Significant missing data was identified in the Volume_Level and Crowd_Size columns.
- **Outliers & Validity:** Impossible values were found, such as negative Volume_Level and Crowd_Size, and Crowd_Energy values exceeding logical bounds

2. Data Cleaning & Preprocessing

To prepare the data for modeling, a rigorous cleaning pipeline was implemented.

2.1 Date and Time Standardization

- **Problem:** The Show_DateTime column contained five distinct formats, including non-standard text descriptions.
- **Solution:** I implemented a function (identify_format) to categorize each entry.
 - **Standardization:** pd.Timestamp to all
 - **Handling Text Descriptions:** Vague terms like "Morning" or "Evening" were mapped to specific reference hours
 - **Feature Extraction:** The timestamp was decomposed into new features: Show_Date_Month, Show_Date_day, and Show_Hour

2.2 Price Normalization

- **Problem:** The Ticket_Price column was a string object containing various currencies and non-numeric characters.
- **Solution:** A cleaning function (ticket_cleaning) was applied to:
 - **Handle "Free":** Explicitly mapped 'Free' entries to 0.0. Later np.nan as scribbles revealed they were just noise.

- **Currency Conversion:** Detected currency symbols (£, €, \$) and applied exchange rates
- **Regex Extraction:** Extracted the base numeric value

2.3 Outlier & Missing Value Handling

- **Strategy:** Negative values in Volume_Level and Crowd_Size (validity issues) were treated as missing values or winsorized.

I didn't use any linear models, because,

- **Limitation:** LR is highly sensitive to outliers. A single data point with an extreme value can disproportionately distort the regression line towards itself
- **Dataset Reality:** As identified in the Data Quality Assessment, the dataset contained significant noise and outlier events, While cleaning mitigated this, residual noise remained. The linear models will not succeed to generalize in the presence of this variance, whereas tree-based models isolated these outliers in specific leaf nodes.

Hyperparameter Tuning Strategy

Methodology: Grid Search with Cross-Validation

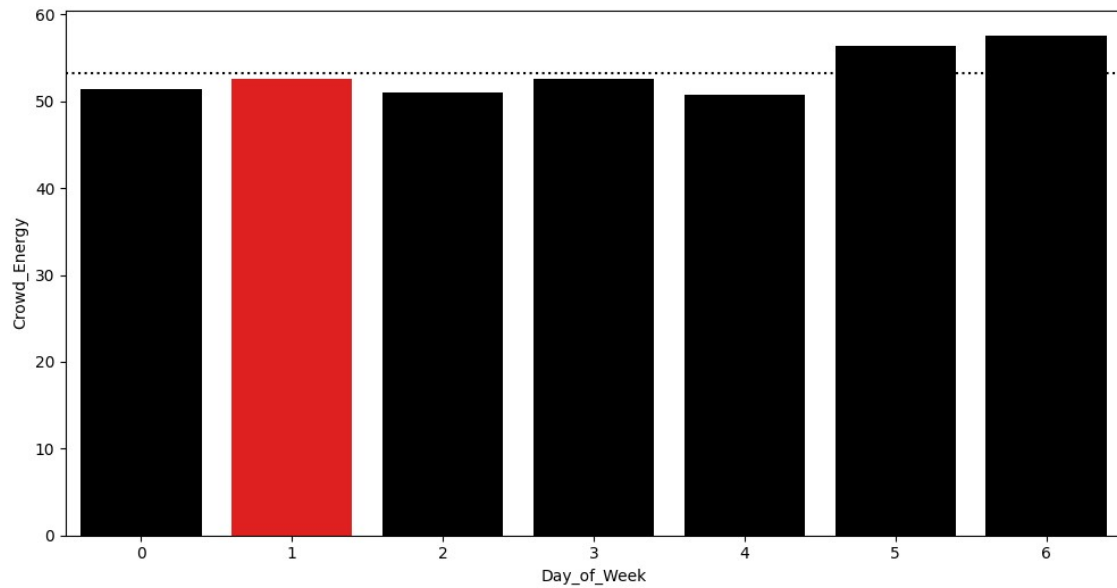
I utilized **GridSearchCV** from the Scikit-Learn library

- **5-Fold Cross-Validation (cv=5):** To ensure the selected parameters generalize well to unseen data, each combination was evaluated 5 times. The data was split into 5 subsets, the model was trained on 4 and validated on the 1 remaining fold. This process was rotated 5 times, and the average score was used.
- **Scoring Metric:** The optimization objective was the **R²** (Coefficient of Determination), prioritizing models that explain the maximum variance in Crowd_Energy.

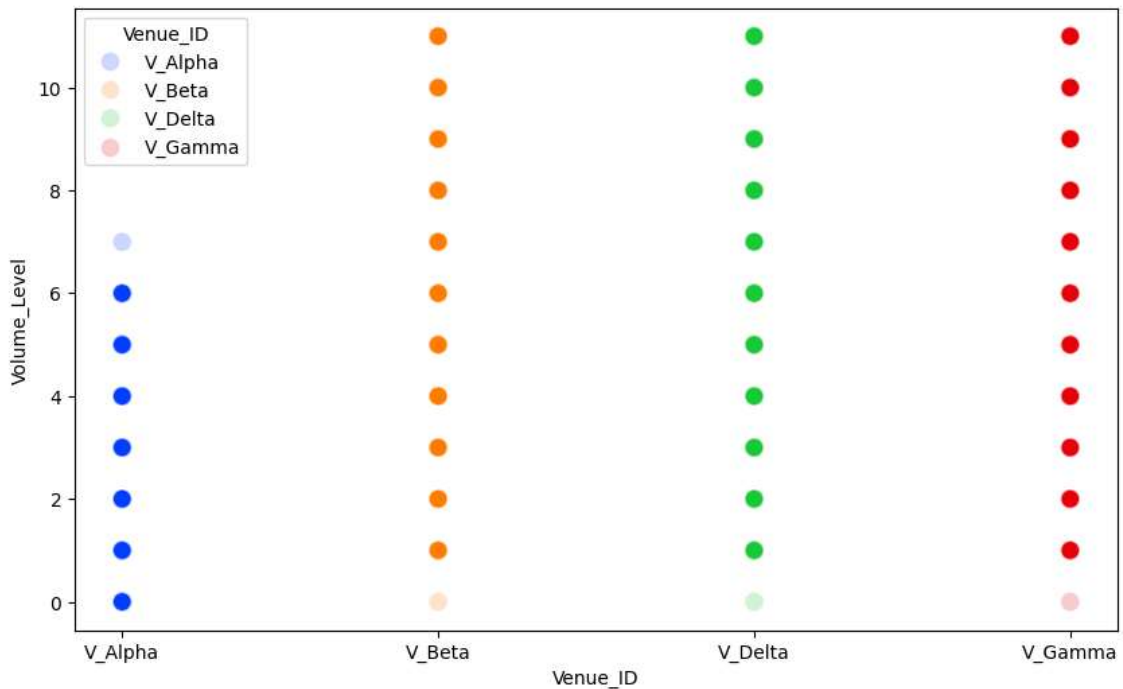
ON SINGER'S CLAIMS

'Tuesday shows are CURSED. I'm absolutely certain of this. Every Tuesday something goes wrong. The amps exploded in Munich (Tuesday). The drummer got food poisoning in Prague (also Tuesday). My voice cracked in Barcelona (you guessed it). I've started refusing Tuesday bookings. Track this in the data - I bet Tuesdays have the lowest energy scores across ALL venues.'

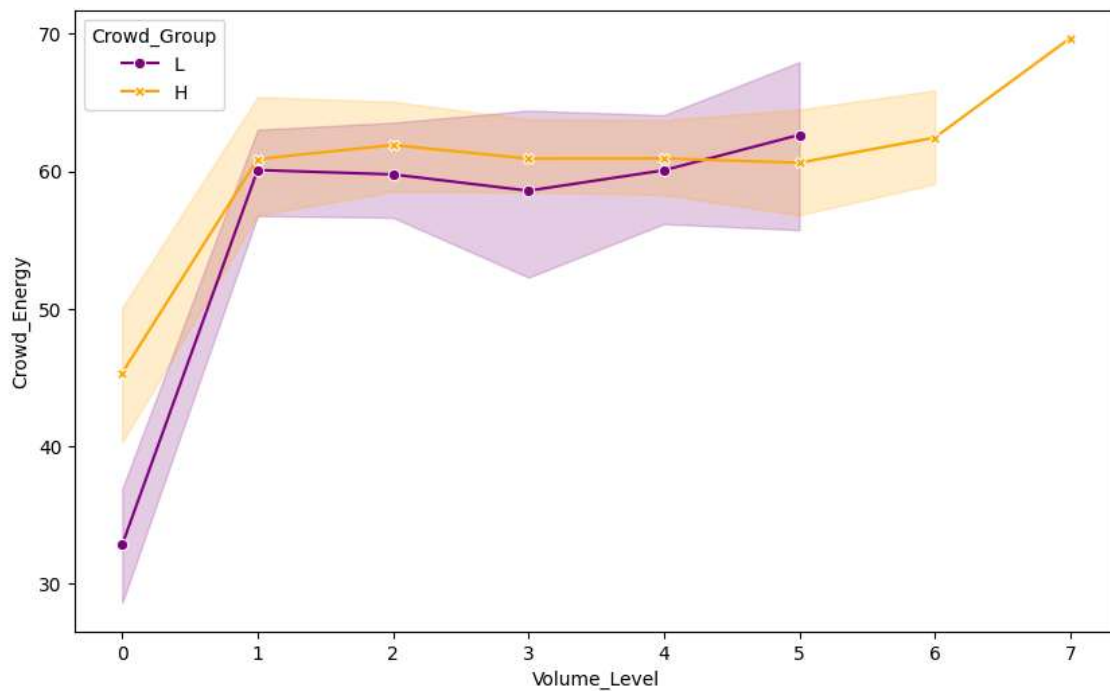
CONCLUSION : This is false, it is below average but not lowest



'The monks at one of the venues... or was it the goths? Someone has a noise limit. Brother Thomas - the old one with the massive beard - he WINCES at a certain volume. Once he winces, it's over. The energy just dies. But here's what's weird: I've ALSO seen him tolerate absolute chaos when the place was packed to the rafters. Maybe he just gives up enforcing when there's too many people? Or maybe he's deaf in one ear and it depends which way he's facing? Or maybe I'm thinking of a completely different venue. The tours blur together, man.'

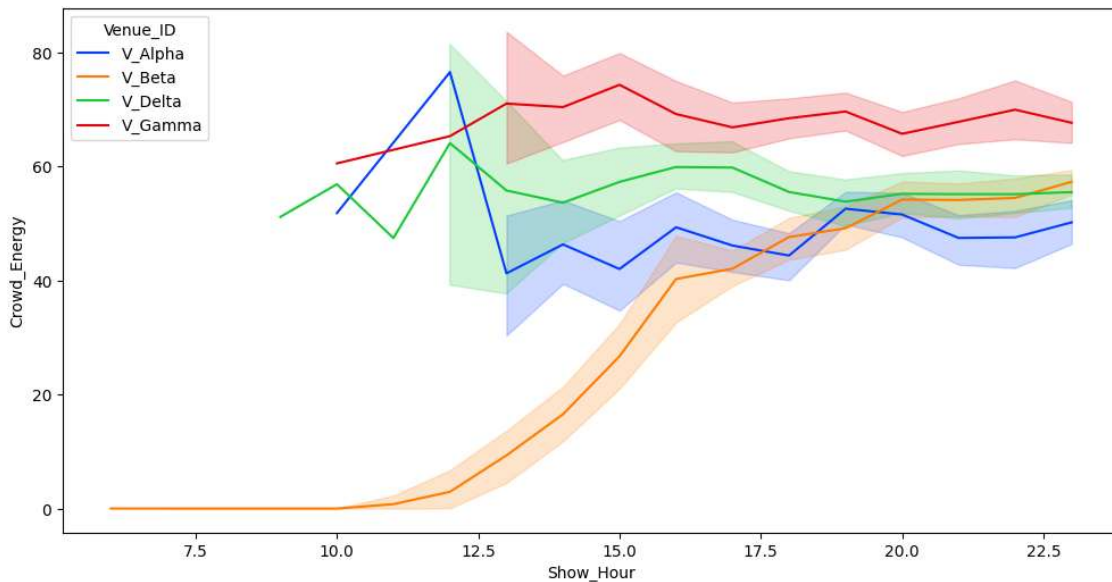


So, she is talking about V_Alpha here, clearly it has a noise limit of 7. Now plotting crowd energy vs vol level to check if he really winces and make the show dead. To check the whole claim i need to make two groups of crowd.



Conclusion:- He most probably winces at volume 5, and also yeah, for higher crowd he goes upto 7 and doesn't wince.

'Speaking of the goths - they ONLY come alive at night. Or wait, is that obvious? Every venue is better at night, right? Actually no, I remember a killer afternoon show somewhere. Was it the mosh pit? The monastery? I genuinely don't remember.'

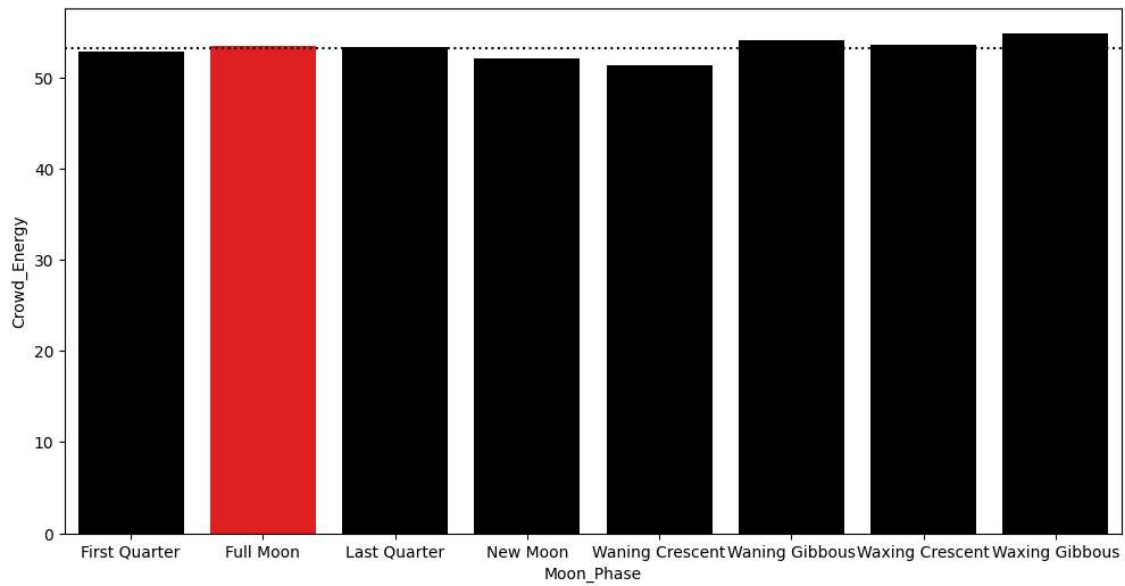


Conclusion:- 'V_Beta' 'the goths' do come at night and it's true, as she has clearly better shows in afternoon in other venues also the killer show was at V_Alpha what she was talking about, the monastery

'FULL MOON = MAGIC 🌕

I am CONVINCED our best shows happen during full moons. The energy is just... different. Electric. The crowd moves like they're possessed.

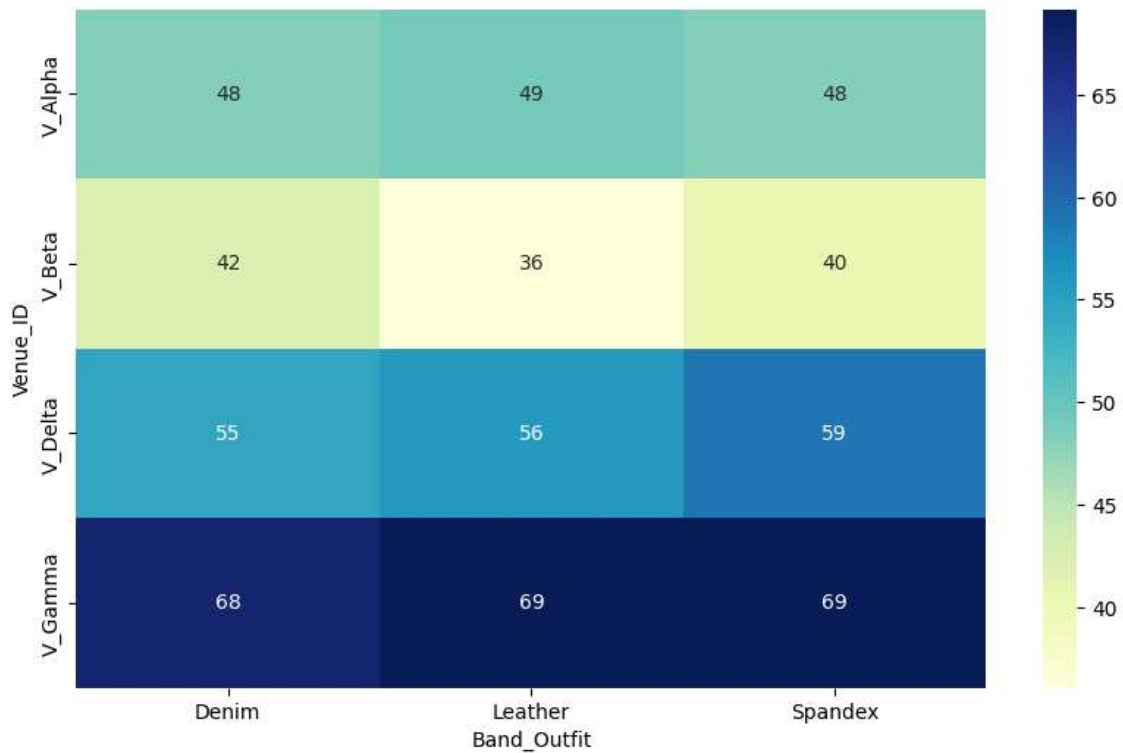
The drummer thinks I'm insane and says it's confirmation bias. But what does he know? He also thinks his lucky socks affect our performance.'



Conclusion:- There is no significant difference between the moon phases

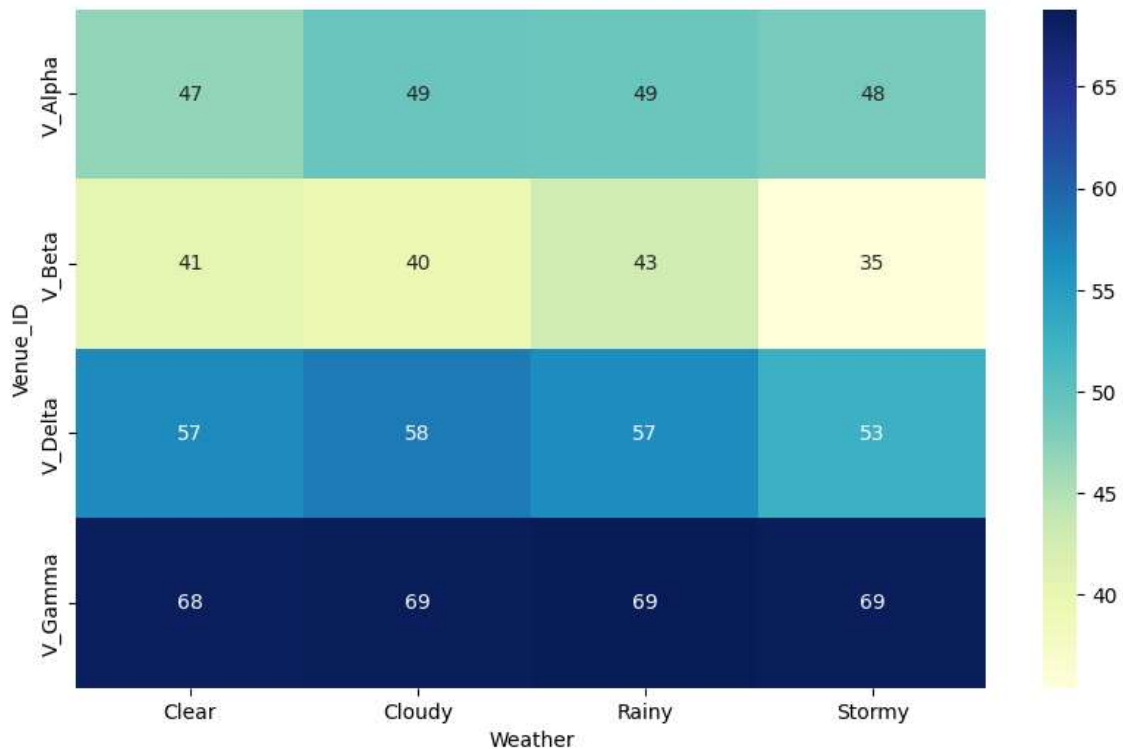
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Actually, speaking of socks - does outfit color matter? I wore red socks in Berlin and they went ABSOLUTELY CRAZY. Coincidence? In Amsterdam I wore the leather jacket and it was mid. But in Rotterdam the leather jacket killed. So maybe it's not about what we wear? Or maybe it is but it interacts with venue? The spandex always FEELS right but the data might say otherwise.'



Conclusion:- No clear evidence of spandex becoming best however it's true wearing leather significantly reduced energy in V_Beta compared to Denim and Spandex

'Also, I've been tracking the weather. Rain definitely sucks. But does it actually affect energy or does it just affect MY mood and I play worse? The outdoor venues (I think V_Delta is partly outdoor?) obviously suffer in storms. Or do they? Mosh pits in the rain are kind of legendary. Ugh, I don't know. Verify this.'



Conclusion:- She is wrong, not V_Delta but V_Beta is worst in storm in V_Delta in storm it's average is less so it maybe partly outdoor and in other places it doesn't matter

'PRICING STRATEGY NOTES (I think this was for the snobs? Or the mosh pit?)

\$45 → they laughed at us. Actually booed. Threw a shoe.

\$55 → mixed reactions, some dancing, some sneering

\$62 → this felt like a sweet spot? Or was this a different venue?

\$80 → definitely worked somewhere. Standing ovation. Or was that the night we had the pyrotechnics malfunction and everyone thought it was part of the show?

The problem is I can't remember WHICH venue had the pricing sensitivity.

I want to say it was V_Gamma because they're snobs. But the mosh pit kids are broke, so maybe THEY'RE the price sensitive ones? Or maybe it's the monastery - monks probably care about value for money?

Also the accountant is USELESS. Half the entries are in pounds, some are in euros, some have dollar signs, and she logs comped tickets as "Free" which throws off all the averages. I once saw an entry for \$500 which is definitely wrong - our max ticket price was like \$100. And there's negative prices in there somehow? Fire her.

EXCHANGE RATES (I looked these up once):

£1 = \$1.27 (approximately)

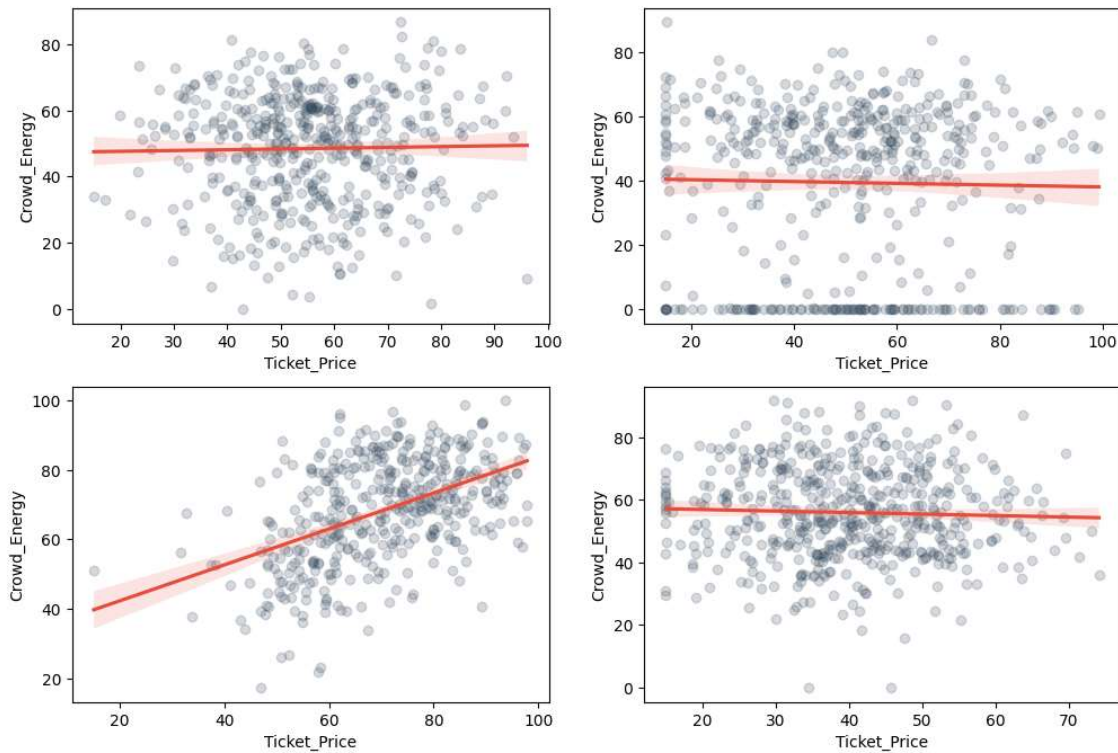
€1 = \$1.09 (approximately)

Actually those might be outdated. Check a converter.'

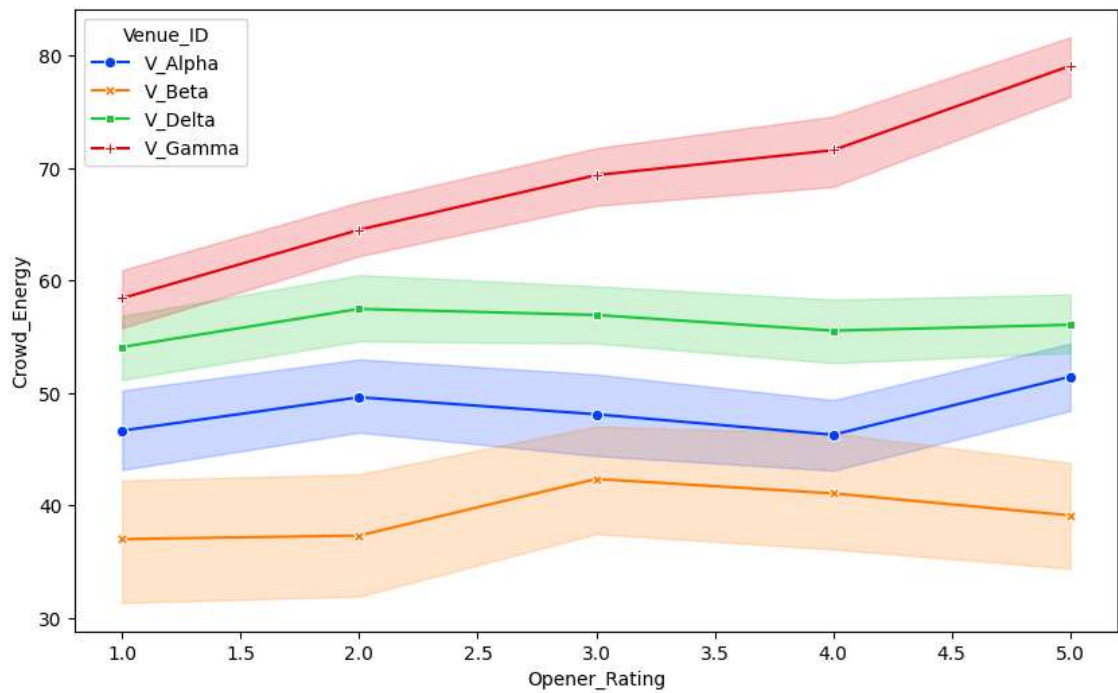
I made the price column fixed already but I missed one thing I took free tickets as 0, which is wrong clearly

Things I THINK are true:

- Volume matters. Somewhere. In some direction.
- Time of day matters. Probably.
- Price matters. To someone.
- Crowd size matters. Obviously? Or not?

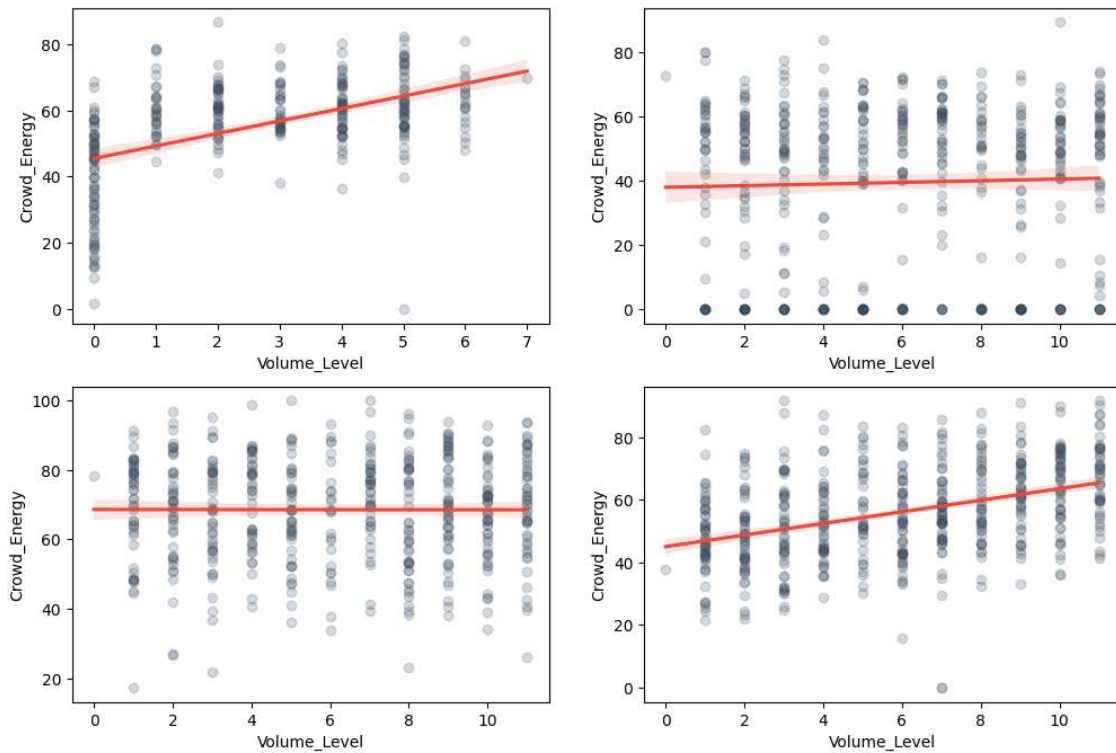


Conclusion:- Clearly In regplot it catches the trend, V_Gamma is only where Ticket_Price matters



Conclusion:- Opener seems to have a linear relationship with opener_rating in V_Gamma

'- Volume matters. Somewhere. In some direction.'



'...is what I used to think. But looking back at the tour logs, there

IS a pattern. The packed shows were INSANE. The empty shows were dead.

But isn't that true everywhere? Maybe V_Delta is special because the

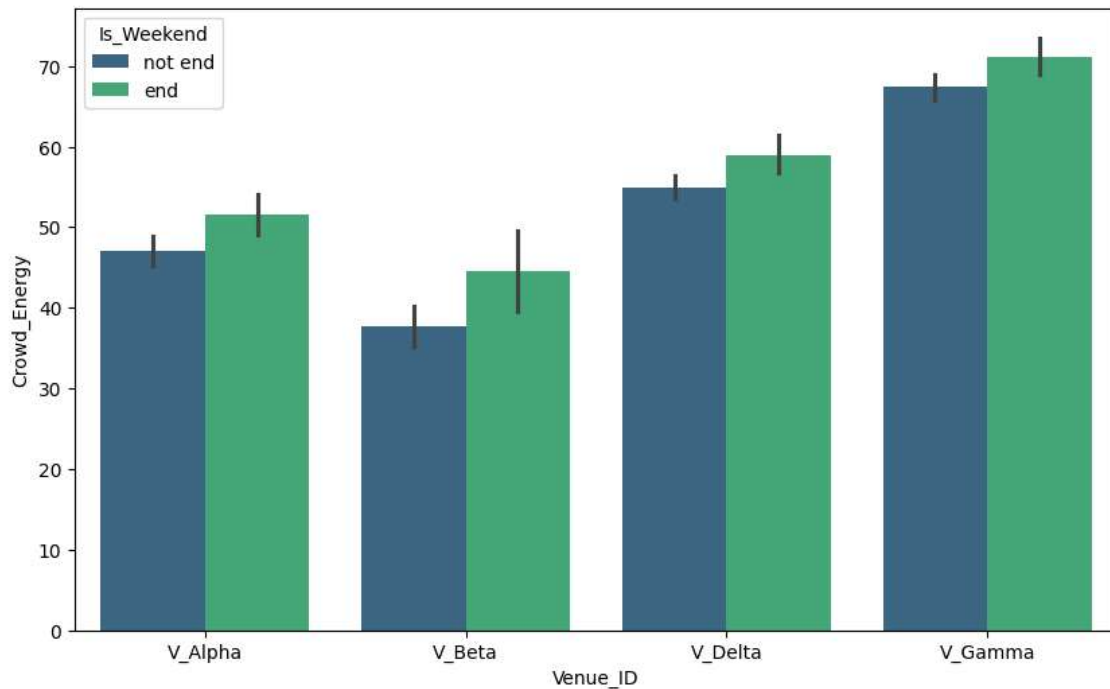
effect is stronger there. The mosh pit FEEDS on itself - more people

means more energy which means more people go crazy which means more

energy. It's exponential. Or linear. Or logarithmic. I don't know math.'

Conclusion: I have plotted crowd energy vs crowd_size above.

Conclusion: I don't think it's much of a factor as there is a proportional trend but slope is v less



Then I tested different tree based model, I tested xgboost, lightgbm, catboost, xtratrees, random forest, hist grad boost, but best result was given by **xgboost and knn imputer**

Then I explored more feature engineering, and got better score with random forest.

Tuned different params using grid search, also tried optuna search for better result

#Venue wise modelling:- They performed worse, because in most of the case the feature which had great correlation with the crowd_energy often had a lot of missing values,

So they failed also less data might be another reason.